

A Review of Coarse-Graining Methods

Insights from Physics, Network Science, and Computer Science

Marco F. da Silva and AccelNet Collabathon participants

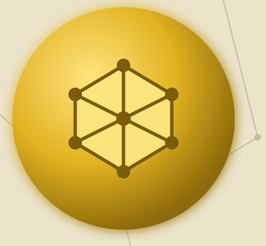
NetSci 2026 · AccelNet-MultiNet Collabathon · CoarseNet



Computer Science



Network Science



Physics

A Review of Coarse-Graining Methods

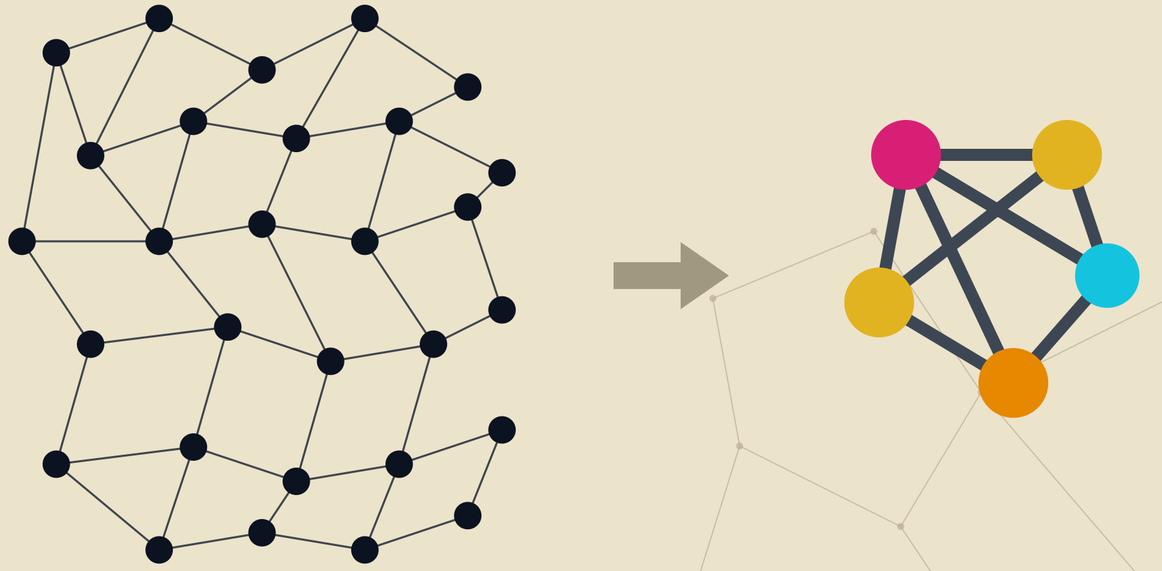
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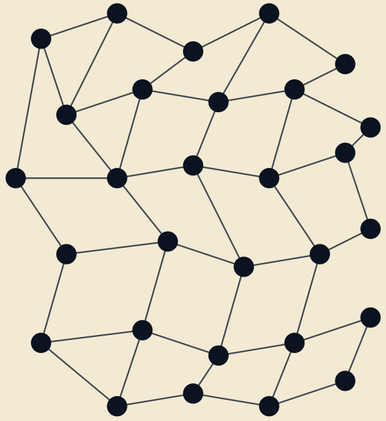
Why Coarse Grain?

- **Reduce** computational complexity for large-scale analysis.
- Identify and **preserve** essential structural properties and dynamics.
- Enable reasoning across **different scales** of the system.



The fundamental engine of coarse-graining is a mathematical choice of mapping and weights.

1 Raw Input



2 Grouping Engine

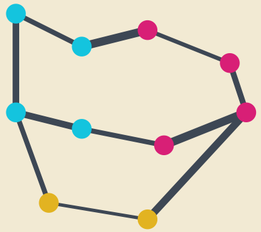
$$f_c : G \rightarrow (M, W) \rightarrow G_m$$

M W

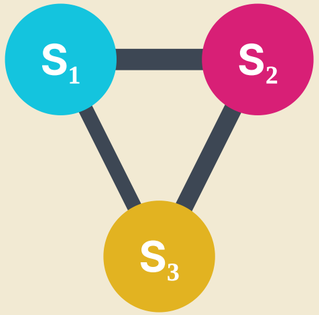
$S_1 \rightarrow n_1, n_2, n_3, n_4$

$S_2 \rightarrow n_5, n_6, n_7, n_8$

$S_3 \rightarrow n_9, n_{10}$



3 Weighted Output



Mere vs. Effective Reduction: The scientific choice of *what to preserve* is logically prior to the choice of the method. Everything downstream is just a flavor of mapping **(M)** and weights **(W)**.

Computer Science prioritizes computational scalability over structural fidelity.

Historical Context

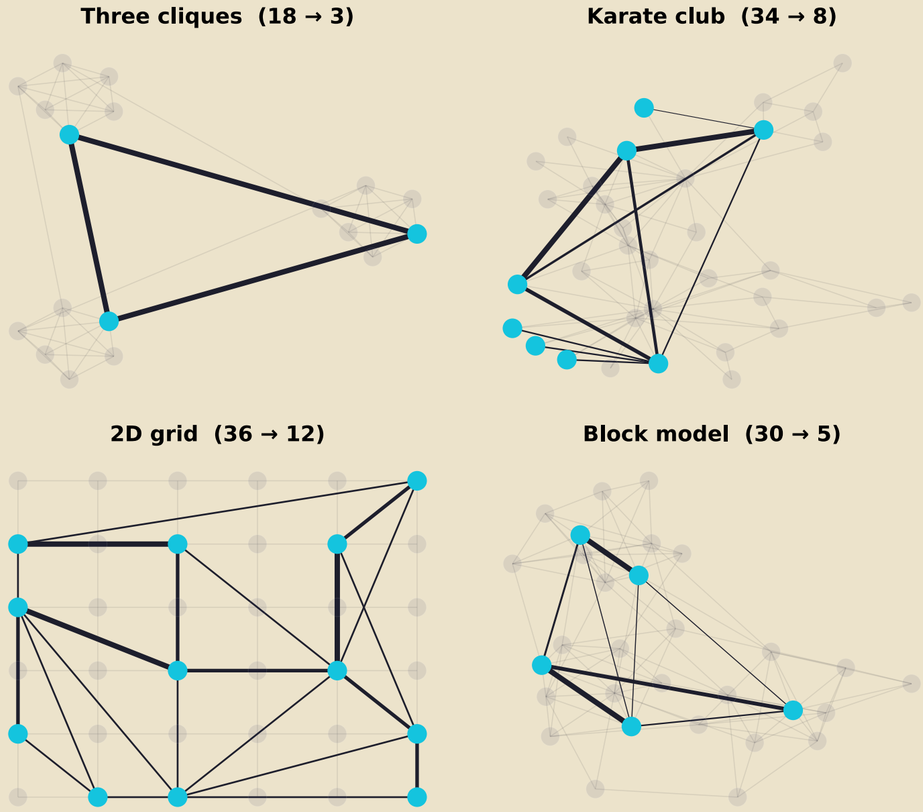
Originating from **multilevel graph partitioning** like Kernighan-Lin, Fiduccia-Mattheyses, and METIS.

Recent Development

MDL-based graph summarization for rapid querying and compression. **Spectral sparsification** with provable guarantees. **Graph condensation** for GNNs.

Core Insight

Scale first, meaning later. The coarse graph does not need to visually or topologically resemble the original system, provided it **computes efficiently**.



Network Science optimizes for the preservation of mesoscale topology and dynamics.

Historical Context

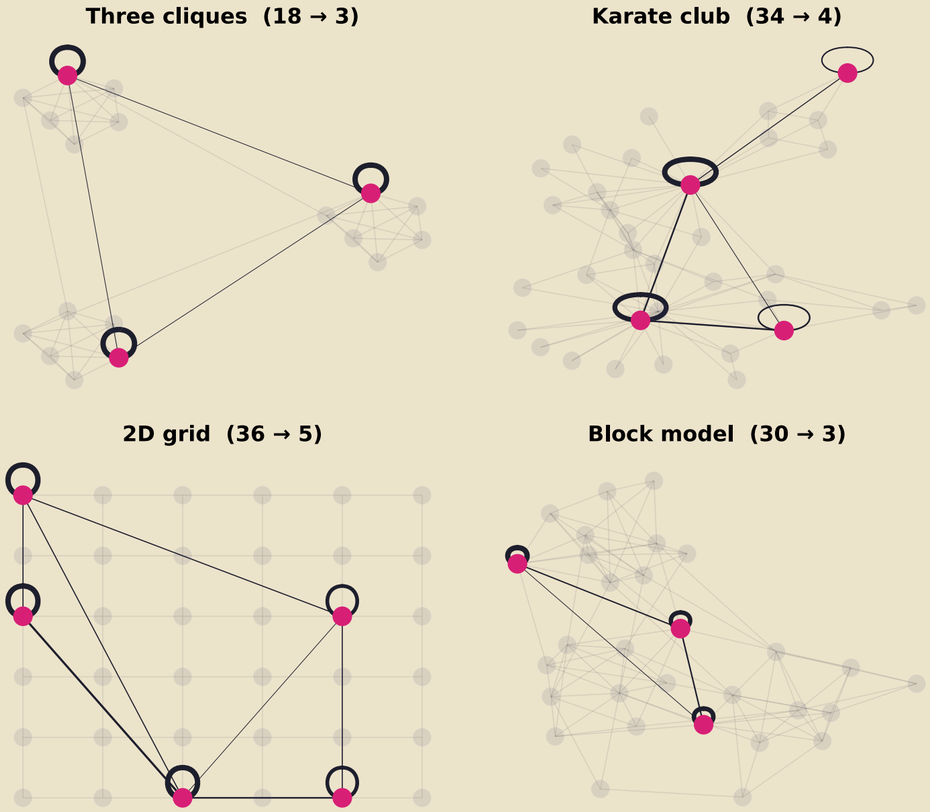
Born from **modularity** and **community detection** in social structures (Girvan-Newman, Louvain, Leiden).

Recent Development

Preservation of specific **structural traits** like degree, spectrum, and motifs. **Dynamics-driven preservation** maintaining random walk behavior or synchronization limits.

Core Insight

Preserve what matters. Methods are strictly tailored to safeguard specific **topological traits** or **dynamical behaviors** through the reduction.



Physics treats coarse-graining as the search for scale invariance and renormalization.

Historical Context

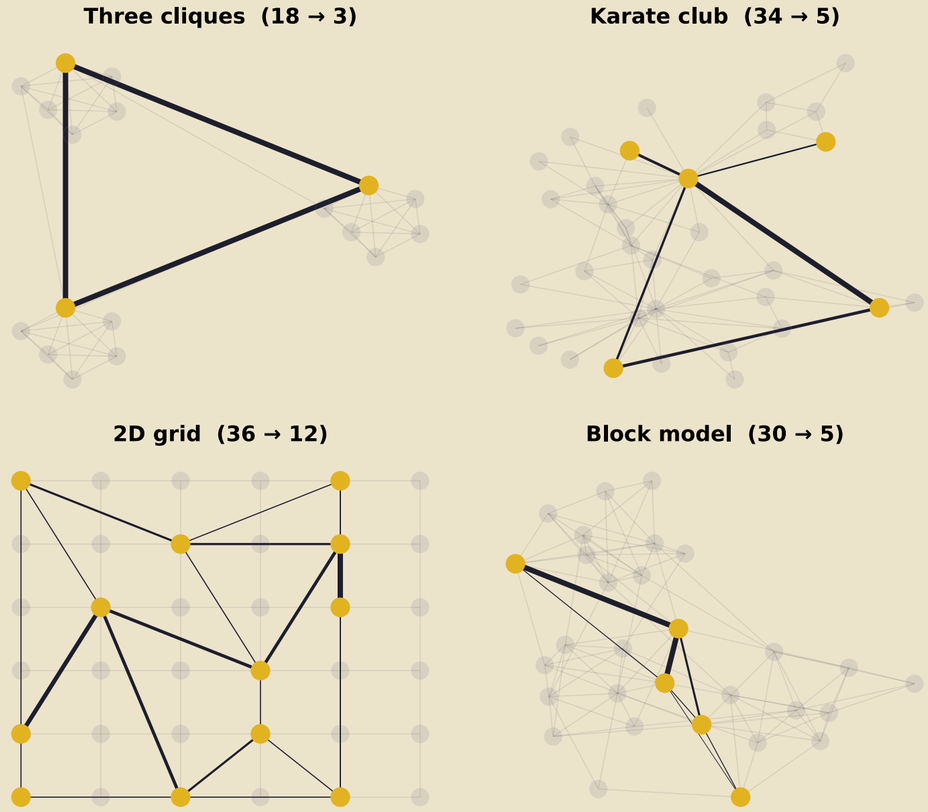
Rooted in the **Renormalization Group** (Kadanoff, Wilson), originally developed to study critical phenomena and **scale invariance** in physical systems.

Recent Development

Box Covering for finding fractal dimensions. **Geometric RG** mapping into latent hyperbolic space. **Laplacian RG** using the diffusion spectrum as a measure of scale.

Core Insight

This is the **most entangled branch**, bridging physics and network science by searching for **thermodynamic-like macroscopic states**.



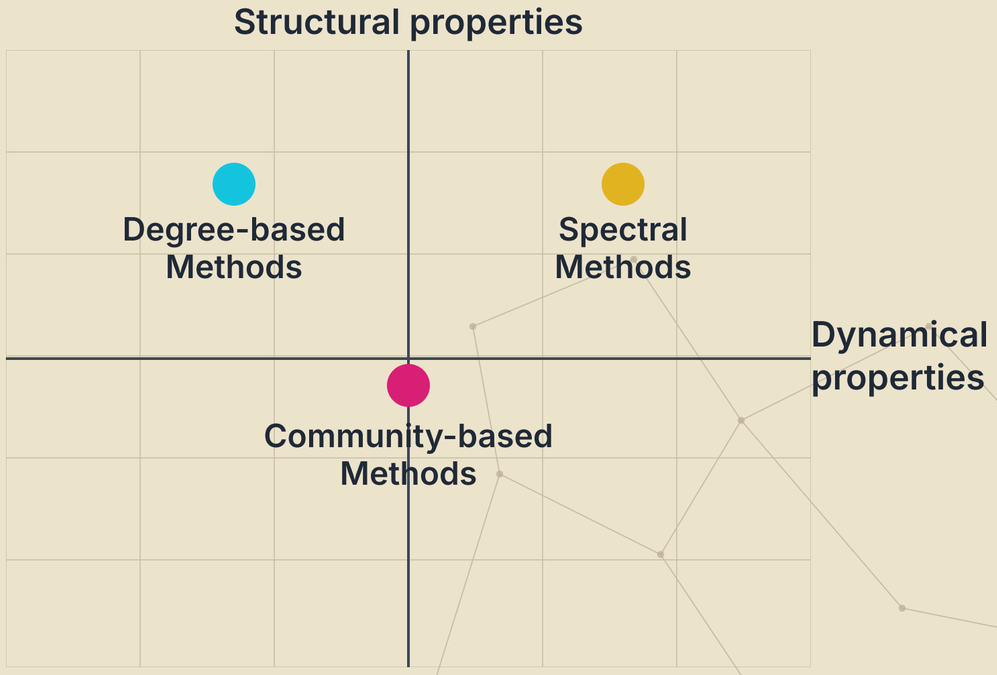
The choice of method fundamentally dictates the scientific reality of the output.

Structural, Dynamical, and Operational Impacts:

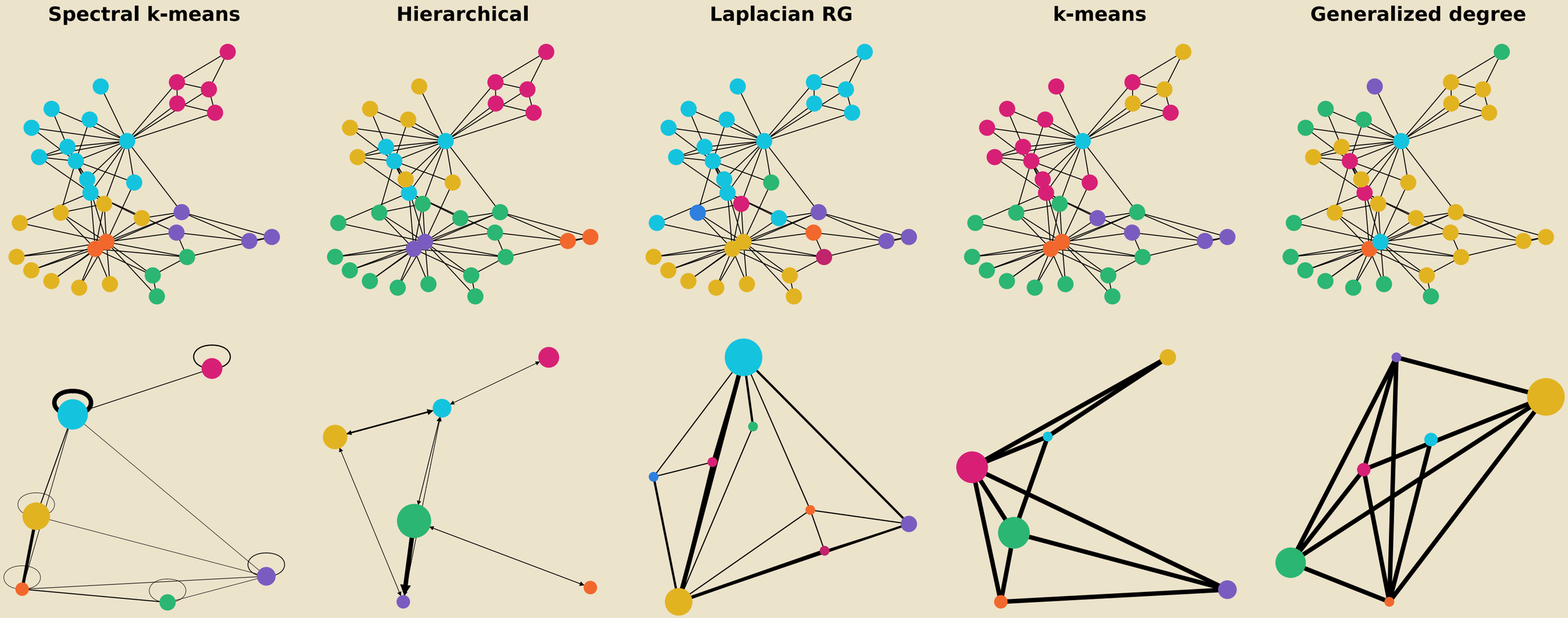
There is no single best method. Preservation is co-defined with the specific scientific question being asked.

The Trade-off:

Spectral methods successfully maintain global structure. Degree-based methods maintain local structure. Community-based methods act as a compromise, landing in the middle of the landscape.



A single input graph yields radically different topologies depending on the chosen lens.



Using the exact same return signature in CoarseNet allows immediate comparison of epistemological frameworks.
method choice = scientific choice.

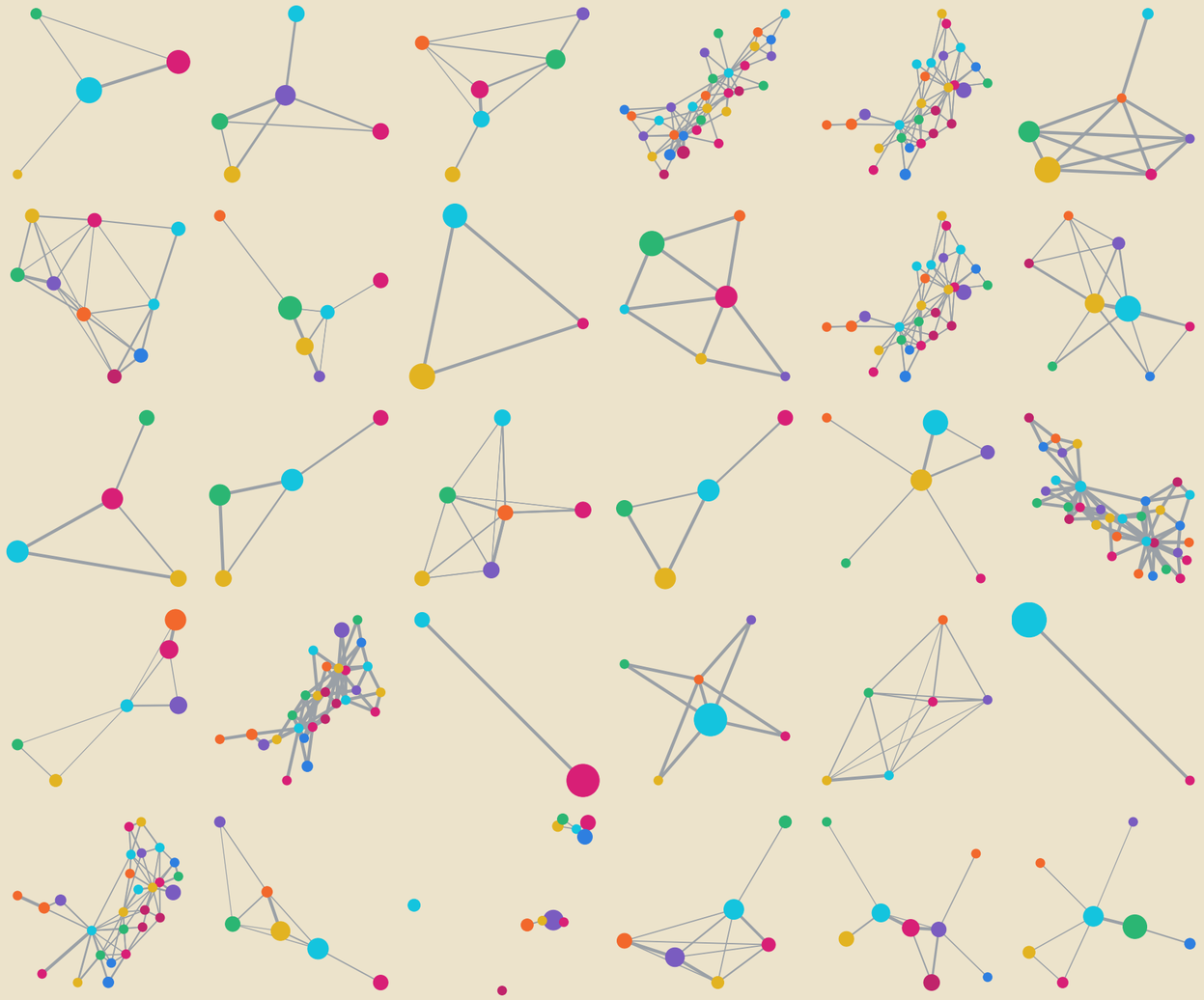
CoarseNet unifies these parallel literatures into a single programmable interface.

```
import coarsenet as cn

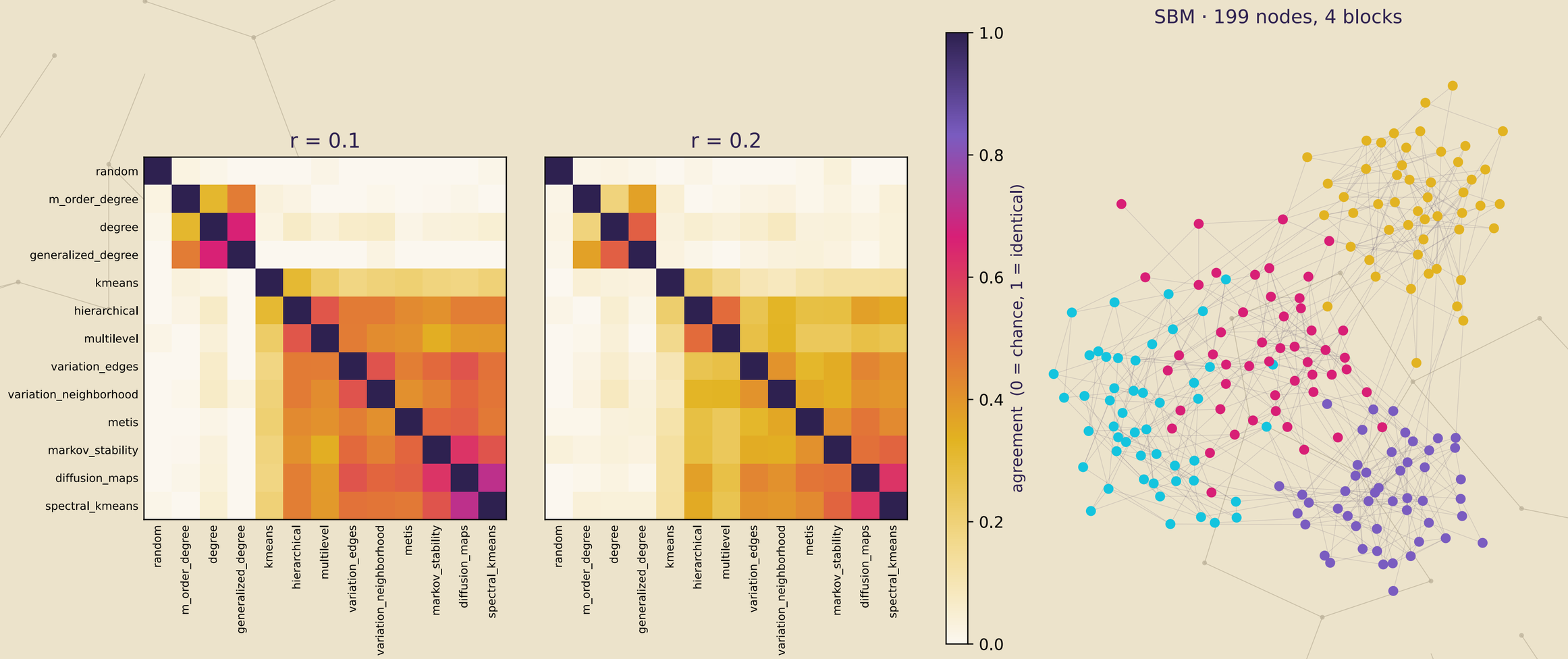
G_cg, mapping = cn.box_covering(G, args)
```

One unified API spanning **Computer Science**, **Network Science**, and **Physics** methods.

30+ algorithms available out-of-the-box. Open Source, pip-installable, and benchmark-ready.



Preliminary results

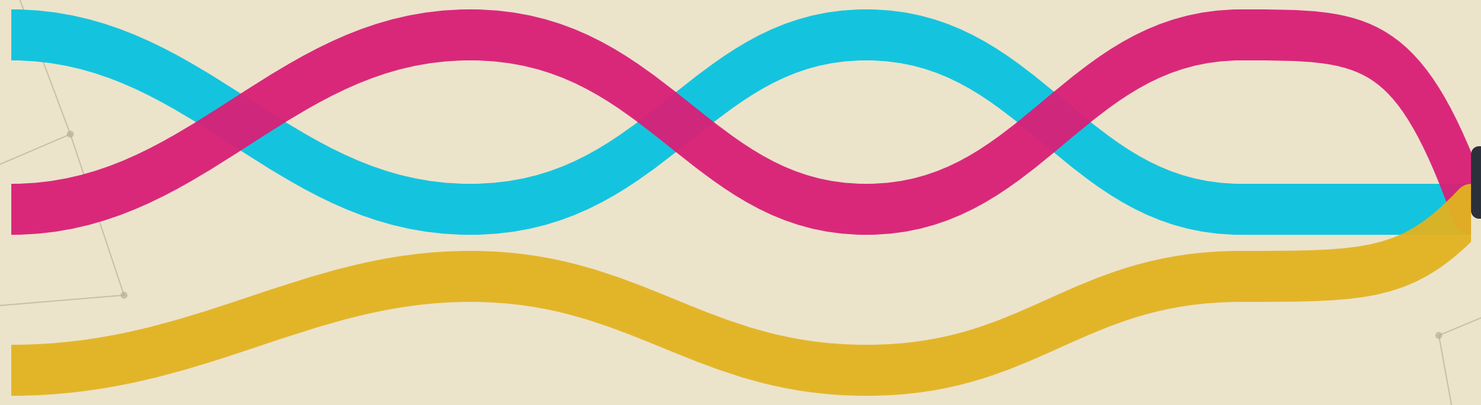


The path forward relies on bridging mathematical equivalences across fields.

Computer Science

Network Science

Physics



CoarseNet

One problem, three fields. Computer Science, Network Science, and Physics have been solving the same coarse-graining problem in isolation.

One unified package. CoarseNet brings **30+ methods** under a single API to test and benchmark them side by side.

The choice is the science. The right method depends on the system and the property you need to preserve.